

36) Poišči maksimum $f(x, y) = 1 + xy$ na enotskem krogu

i) eksplicitno

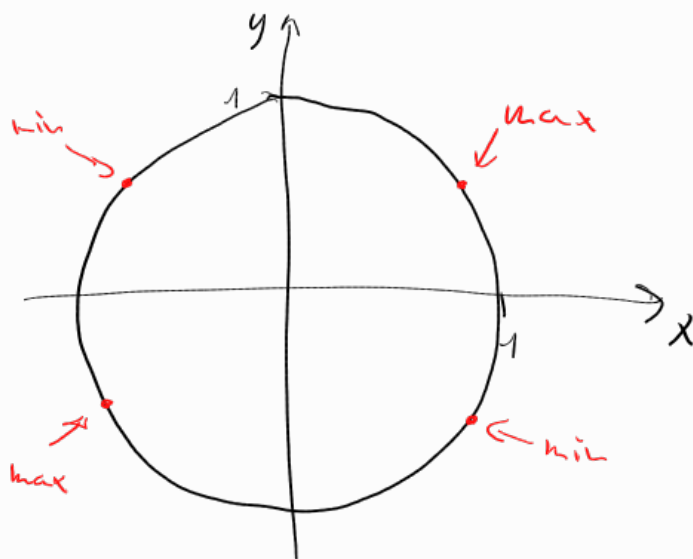
$$y(x) = \pm \sqrt{1-x^2}$$

$$\rightarrow f(x, y(x)) = 1 + x\sqrt{1-x^2}$$

$$\frac{\partial f}{\partial x} = +\sqrt{1-x^2} + \frac{x}{2} \frac{1 \cdot (-2x)}{\sqrt{1-x^2}} = 0$$

$$1-x^2 - x^2 = 0$$

$$x = \pm \frac{\sqrt{2}}{2}$$



ii) Lagrangeovi multiplikator

vez: $g(x, y) = x^2 + y^2 - 1 = 0$

• iščemo točko, v kateri velja $\nabla f \parallel \nabla g$

$$\mathcal{L}(x, y, \lambda) = f(x, y) + \lambda g(x, y)$$

$$\mathcal{L}_x = \mathcal{L}_y = \mathcal{L}_\lambda = 0$$

$$\nabla f = -\lambda \nabla g$$

$$\nabla f = (y, x)$$

$$\nabla g = (2x, 2y)$$

$$\left. \begin{array}{l} y = -\lambda \cdot 2x \\ x = -\lambda \cdot 2y \end{array} \right\} \rightarrow x - y = -2\lambda(y - x)$$

$$x = -\lambda \cdot 2y$$

$$x^2 + y^2 - 1 = 0$$

$$x^2 + x^2 = 1$$

$$x = \pm \frac{1}{\sqrt{2}}$$

$$\lambda = \frac{1}{2}$$

ni reš. (min)

$$x = y$$

iii) parametrična rešitev

$$x = \cos \varphi, y = \sin \varphi, \varphi \in [0, 2\pi)$$

$$f(\varphi) = 1 + \sin \varphi \cos \varphi = 1 + \frac{1}{2} \sin 2\varphi$$

$$f'(\varphi) = \cos 2\varphi = 0 \rightarrow 2\varphi = \frac{\pi}{2} + k\pi \rightarrow \varphi_1 = \frac{\pi}{4}, \varphi_2 = \frac{3\pi}{4}, \varphi_3 = \frac{5\pi}{4}, \varphi_4 = \frac{7\pi}{4}$$

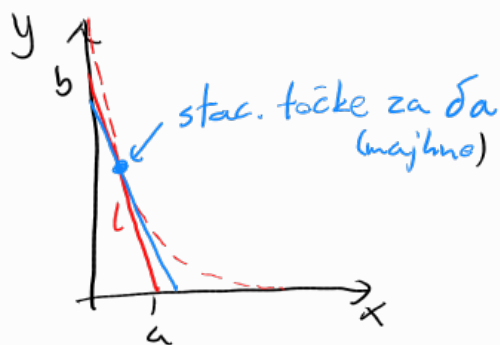
max

min

max

min

3.7) Enačba ovjnice tetiv dolžine l med ordinato in absciso



$$y = b - \frac{b}{a}x$$

$$\frac{y}{b} + \frac{x}{a} = 1, \quad a^2 + b^2 = l^2$$

$$\frac{y}{\sqrt{l^2 - a^2}} + \frac{x}{a} = 1 \quad g(x, y, a) = \frac{y}{\sqrt{l^2 - a^2}} + \frac{x}{a} - 1 = 0$$

$$g_a = y \frac{a}{(l^2 - a^2)^{3/2}} - \frac{x}{a^2} = 0 \quad \dots \quad a = \frac{x^{1/3} l}{\sqrt{y^{2/3} + x^{2/3}}}, \quad b = \frac{y^{1/3} l}{\sqrt{x^{2/3} + y^{2/3}}}$$

$$\rightarrow \frac{y}{b} + \frac{x}{a} = 1 \rightarrow \frac{y \sqrt{x^{2/3} + y^{2/3}}}{y^{1/3} l} + \frac{x \sqrt{x^{2/3} + y^{2/3}}}{x^{1/3} l} = \frac{(x^{2/3} + y^{2/3}) \sqrt{x^{2/3} + y^{2/3}}}{l} = 1$$

$$\rightarrow \boxed{x^{2/3} + y^{2/3} = l^{2/3}} \text{ Astroida}$$

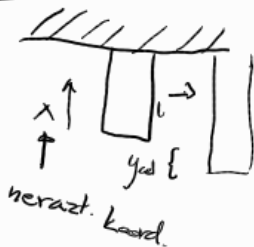
3.1 Kodre 30/9

$$l = 10 \text{ m}$$

$$E = 500 \text{ N/cm}^2$$

$$\rho = 1,5 \text{ g/cm}^3$$

$$y_{\text{cel}} = ? \text{ (rast. zar. last. točka)}$$



$$F(x+dx) - F(x) = g dm = g \cdot \rho \cdot S dx$$

$$\rightarrow F(x) = \rho g S x$$

$$\frac{F}{S} = E \frac{dy}{dx} \quad \leftarrow \text{rast. zar.}$$

$$\int_0^{y_{\text{cel}}} \rho g x dx = \int_0^{y_{\text{cel}}} E dy$$

$$y_{\text{cel}} = \frac{\rho g l^2}{2E} = \underline{\underline{0,15 \text{ m}}}$$

Kodre 30/11 Epruveta s plinom

$$\rho(r) = ?$$



$$F_c = m\omega^2 r$$

$$dF_c = \omega^2 r dm = \omega^2 r \rho S dr$$

$$pV = \frac{m}{M} RT$$

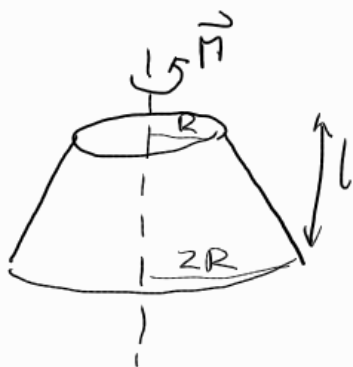
$$dp = \frac{RT}{M} d\rho$$

$$dp = \frac{dF}{S} = \omega^2 \rho r dr$$

$$\int_{\rho_0}^{\rho} \frac{RT}{M} \frac{d\rho}{\rho} = \int_0^r \omega^2 r dr = \frac{\omega^2 r^2}{2}$$

$$\ln \frac{\rho}{\rho_0} = \frac{M\omega^2 r^2}{2RT} \rightarrow \underline{\rho(r) = \rho_0 \exp\left(\frac{M\omega^2 r^2}{2RT}\right)}, \quad \rho(r) = \rho_0 \exp\left(\frac{M\omega^2 r^2}{2RT}\right)$$

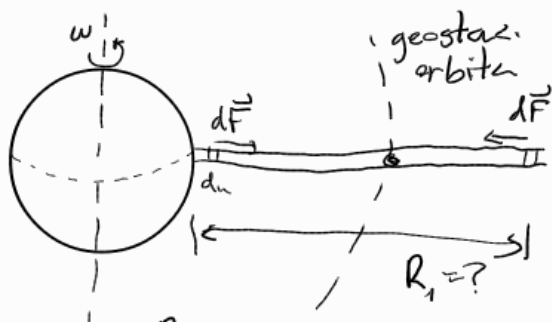
3.4



$$\frac{F}{S} = G \frac{\Delta l}{l}, \quad \text{sp. ploškar je fiksna, in zg. delujeno z navoreni}$$

$$\Delta \varphi = ?$$

3.5 Vesoljsko dvigalo



$$dF = dF_c - dF_g = d(m\omega^2 r) - d\left(\frac{mMG}{r^2}\right) =$$

$$= \omega^2 r dm - \frac{MG}{r^2} dm =$$

$$= \left(\omega^2 r - \frac{MG}{r^2}\right) S \rho dr$$

$$F = \int_{r_2}^{r_1} \underbrace{\left(\omega^2 r - \frac{MG}{r^2}\right) S \rho dr}_{-\nabla V(r)} = \rho S \left(V(r_2 + l) - V(r_2) \right) =$$

$$= \rho S \left(-\frac{\omega^2}{2} (R_1^2 - r_2^2) - MG \left(\frac{1}{R_1} - \frac{1}{r_2} \right) \right) =$$

$$V(r) = -\frac{\omega^2 r^2}{2} - \frac{MG}{r} \Big|_{r_2}^{R_1} = 0$$

$$r_1 \gg r_2$$

$$\frac{Mg}{r_2} = \frac{\omega^2 R_1^2}{2} \rightarrow R_1 = 1,6 \cdot 10^8 \text{ m}$$